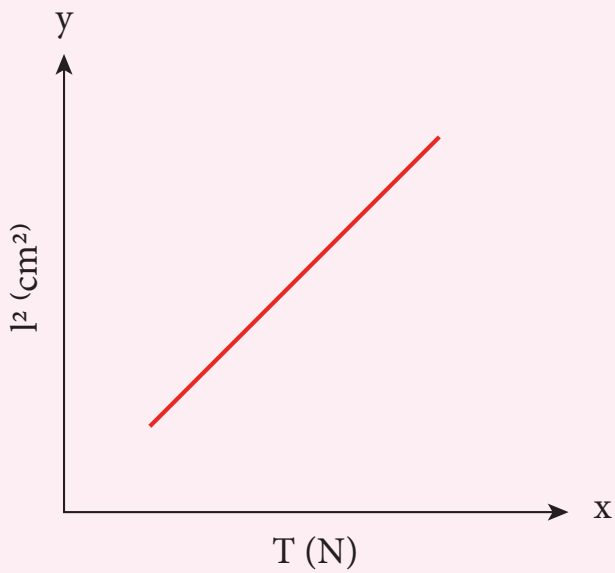
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PROCEDURE

- Set up the sonometer on the table and clean the groove on the pulley to ensure that it has minimum friction.
- Set a tuning fork of known frequency into vibration by striking it against the rubber pad. Plug the sonometer wire and compare the sound due to the vibration of tuning fork and the plugged wire.
- Adjust the vibrating length of the wire by the adjusting the bridge B till the two sounds appear alike.
- Place a mass of 1 kg for initial reading in the load hanger.
- For final adjustment place a small paper rider R in the middle of the wire AB.
- Now, strike the tuning fork and place its shank stem on the bridge A and then slowly adjust the position of the bridge B till the paper rider is agitated violently (might eventually falls) indicating resonance.
- Measure the length of the wire between wedges at A and B which is the fundamental mode corresponding to the frequency of the tuning fork.
- Increase the load on the hanger in steps of 0.5 kg and each time find the resonating length as done before with the same tuning fork.
- Record the observations in the tabular column.

MODEL GRAPH



Relation between tension T (N) and length l^2 (cm^2)